

Monte Carlo Simulations versus Historical Simulations

A second classical approach to studying retirement withdrawal rates is to use Monte Carlo simulations that are parameterized to the same historical data used in historical simulations. This can be done either by randomly drawing past returns from the historical data to construct thirty-year sequences of returns (a process known as “bootstrapping”), or by simulating returns from a statistical distribution (usually a multivariate normal or lognormal distribution) that matches the historical parameters for asset returns, standard deviations, and correlations. Simply put, Monte Carlo simulations provide a way to develop sequences of random market returns fitting predetermined characteristics, in order to test how financial plans will perform in a wider variety of good and bad market environments.

Financial planner Lynn Hopewell implored the financial advisory profession to adopt Monte Carlo simulation tools in his seminal 1997 article in the *Journal of Financial Planning*. He argued forcefully against merely developing spreadsheets for financial plans based on average input assumptions or by testing the robustness of plans with worst-case scenarios. He said those approaches do not provide probabilities of outcomes, and it is still difficult to know how to develop a financial plan, save, and spend using such limited analysis. Today, Monte Carlo analysis is at the heart of most financial planning software programs that aim to test the feasibility of financial plans.

Monte Carlo simulations have a number of advantages over their historical simulations counterparts used in the analysis for Bengen’s work and the Trinity study. First, Monte Carlo allows for a wider variety of scenarios than the rather limited historical data can provide. Between 1926 and 2016, there are only sixty-two rolling thirty-year periods, and these are not independent periods since they share so many overlapping data points. Meanwhile, it is not uncommon to see a Monte Carlo simulation study based on ten thousand or more simulated paths for financial market returns. This provides an opportunity to observe a wider variety of return sequences that support a deeper perspective about possible retirement outcomes.